Exponential Applications 2

1. **Annuity**

An **annuity** is a financial contract that provides a series of regular, fixed payments over a specific period.

Depending on your goal, an annuity functions in one of two ways:

- **Accumulation Phase (Building):** You make regular contributions to an account that grows with interest over time to reach a future goal.
- **Distribution Phase (Receiving):** You start with a lump sum that is paid out to you in periodic installments until the balance reaches zero.

1.1. Building An Annuity - Future Value

Building an annuity.

Invest | deposit \$1000 every month (in the end of the month) at month 4 interest of 10% for 6 months.

How much the investment is worth in the end of 6 months?

The Formula

$$FV = P \times \left[\frac{(1+r)^n - 1}{r}\right]$$

Where:

- FV: The Future Value (the total amount you want to have).
- P: The Periodic payment (the amount you deposit each period).
- r: The interest rate per period (expressed as a decimal).
- n: The total number of periods (e.g., total years or months).

In the example: $P = \$1000$

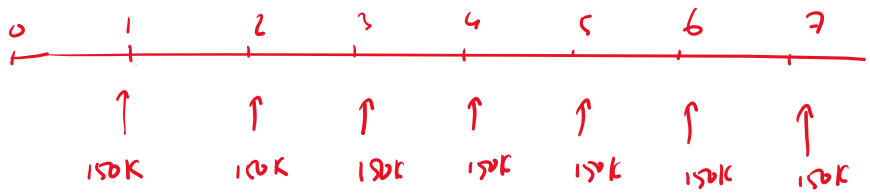
$r = .1$

$n = 6$

$$FV = 1000 \cdot \left[\frac{(1+.1)^6 - 1}{.1}\right] = 7,715.61$$

Example: A rookie player in the NFL just signed his first 7-year contract. To prepare for his future, he deposits $150,000 at the end of each year for 7 years in an account with 3.1% compounded annually.

a. How much will he have in his account after 7 years?



$P = 150,000$

$r = 3.1\%$

$n = 7$

$$FV = P \cdot \left[\frac{(1+r)^n - 1}{r} \right] = 150,000 \cdot \left[\frac{1.031^7 - 1}{.031} \right]$$

$$= 1,152,834.59$$

b. How much interest was earned over the 7 years?

Total deposits : $150,000 \times 7$

Interest earned : $1152834.59 - 150,000 \times 7$

$= 102,834.59$

You try: A new professional basketball player signs a 6-year contract. To save for the future, he deposits $120,000 at the end of each year into an investment account that earns 3.4% interest compounded annually.

a. How much money will be in the account after 6 years?

b. How much interest was earned during the 6 years?

1.2. Receiving An Annuity- Present Value

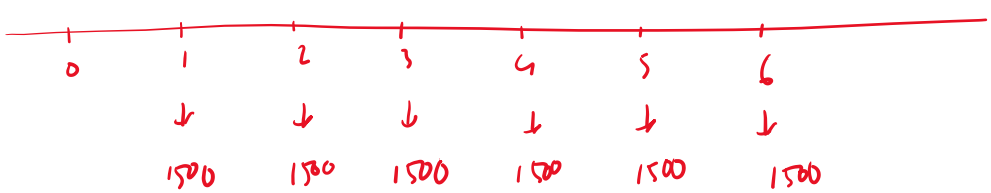
The Formula

$$PV = P \times \left[\frac{1-(1+r)^{-n}}{r}\right]$$

Where:

- PV: The Present Value (the lump sum you start with today).
- P: The Periodic payment (the amount you receive each period).
- r: The interest rate per period (as a decimal).
- n: The total number of periods.

Example : A wealthy family member has funded an annuity that will pay you $1500 at the end of each year for six years. If the interest rate is 8%, compounded annually, find the present value of the annuity.



$$PV = P \cdot \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

$P = 1500$

$r = .08$

$n = 6$

$$\Rightarrow \quad PV = 1500 \times \left[\frac{1 - (1 + .08)^{-6}}{.08} \right]$$

$$= 6934.319$$

You try: A relative has set up an annuity that will pay you $2,000 at the end of each year for 5 years. The account earns 7% interest compounded annually. Find the present value of the annuity today.

Example: To supplement his pension in the early years of his retirement, Ryan plans to use $124,500 of his savings as an ordinary annuity that will make monthly payments to him for 20 years. If the monthly interest rate is 2.8%, how much will each payment be?

$$PV = P \times \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

$n = 20 \times 12 = 240$

$r = 2.8\%$

$PV = 124,500$

Find P

$$124,500 = P \cdot \left[\frac{1 - (1 + .028)^{-240}}{.028} \right]$$

$$124,500 = P \times 35.66$$

$$\Rightarrow \quad P = 3490.619$$

You try: To fund his extensive pickleball tournament travel and new paddle upgrades for the next 15 years, Sarah decides to invest $\$150,000$ into an ordinary annuity. If the annuity offers an annual interest rate of 3.5% compounded monthly, how much will Sarah receive in monthly payments?

2. Loans and Amortization

Amortization is the process of paying off a debt over a set period through a series of regular, fixed payments.

$$M = L \times \left[\frac{r(1+r)^n}{(1+r)^n - 1} \right]$$

Variable Guide

- M: Monthly Payment (The amount you pay each period).
- L: Loan Amount (The total principal you are borrowing today).
- r: Monthly Interest Rate (Annual Rate / 12).
- n: Total Number of Payments (Loan Term in years $\times$ 12).

Example: The average used car price is $32,000. The average annual interest rate is currently 6.39% for a 60-month loan (5 years). If you have $5,000 for a down payment,

a. what can you expect the monthly payment to be?

$$M = L \times \left[\frac{r(1+r)^n}{(1+r)^n - 1} \right]$$

$L = 32{,}000 - 5000 = 27{,}000$

$n = 60$

$r = 6.39\% / 12 = .005325$

$$M = 27000 * \left[\frac{.005325 * (1 + .005325)^{60}}{(1 + .005325)^{60} - 1} \right]$$

$= 526.896$

b. How much will you pay in total for the car?

526.896 * 60 = 31,613.76

You try 1: The average rate for a 30-year fixed mortgage is 7.5%. Assume a down payment of 20% on a home purchase of $315,000.

a. Find the monthly payments needed.

b. How much will be paid in total over the life of the loan?

You try 2: In 2023, the average amount of student loan debt for a bachelor's degree recipient at a private university was $54,921. Assume an interest rate of 6.87%

a. Find the monthly payment and calculate the total interest paid over the course of the loan for a 10-year payment plan.

b. Find the monthly payment and calculate the total interest paid over the course of the loan for a 20-year payment plan.